rv is constant on the union of all those finite sets. The argument also depends on the preceding corollary after its parameter and multidimensional gaps are repaired. A shorter alternative is the human referee's generic-lift argument: over a model containing the parameters, a tuple of full RV-dimension has a lift realizing one complete type, so its lift cannot meet A partially; state the dimension/type fact precisely if using this route.

Author resolution. The author approves choosing exactly one VF representative of each boundary RV tag and stating constancy of the finite RV-image set along each source RV fiber. The proof now splits zero-coordinate faces before applying Corollary 2.30, retains all input tags in the graphs, and gives the zero-dimensional base case and the coordinate-chain coverage argument.

Let $f : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ be a definable function.

Lemma 2.36 ([17, Lemma 8.7])**.** *There is a definable closed set $A \subseteq \mathrm{VF}^n$ with $\dim_{\mathrm{VF}}(A) < n$ such that $f \restriction (\mathrm{VF}^n \smallsetminus A)$ is continuous.*

Per the usual terminology, this lemma says that f is continuous *almost everywhere*.

Definition 2.37. In this definition, for a nonzero tuple $x \in \mathrm{VF}^k$ we set $\mathrm{val}(x) = \min_i \mathrm{val}(x_i)$, and we set $\mathrm{val}(0) = \infty$. This minimum convention is separate from the coordinatewise valuation image used in Terminology 2.39.

For any $a \in \mathrm{VF}^n$, we say that f is *differentiable at a* if there is a linear map $\lambda : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ (of VF-vector spaces) such that, for any $\epsilon \in \Gamma$, if $b \in \mathrm{VF}^n \setminus \{0\}$ and $\mathrm{val}(b)$ is sufficiently large then

$$\mathrm{val}(f(a+b) - f(a) - \lambda(b)) - \mathrm{val}(b) > \epsilon.$$

It is straightforward to check that if such a linear function λ (represented by a matrix with entries in VF) exists then it is unique and hence is called the *derivative of f at a*.

Write $f = (f_1, \ldots, f_m)$. For $a = (a_i, a_{\tilde{i}}) \in \mathrm{VF}^n$, if the derivative of the function $f_j \restriction (\mathrm{VF} \times a_{\tilde{i}})$ at a_i exists then we call it the *ijth partial derivative of f at a*. Whenever f is differentiable at a, the (j,i)-entry of its derivative matrix is this ijth partial derivative.

Resolved referee R034 [Clarification; revised] – Valuation of a tuple and the zero increment. *Original source line 980.* Define $\mathrm{val}(b) = \min_i \mathrm{val}(b_i)$ for a nonzero tuple, with $\mathrm{val}(0) = \infty$, and require $b \neq 0$ in the difference quotient. Otherwise the display includes the undefined expression $\infty - \infty$ at $b = 0$. Later, where valuations of tuples are taken coordinatewise (Terminology 2.39), distinguish that use from this minimum convention. The input-first indexing of partial derivatives is legitimate, but please keep the corresponding matrix convention explicit.

Author resolution. The author approves the minimum coordinate valuation for differentiability, with $\mathrm{val}(0) = \infty$, and excludes the zero increment from the difference quotient. This avoids $\infty - \infty$. The input-first ijth partial-derivative convention is retained and its derivative-matrix position (j,i) is explicit. The separate boundedness convention remains coordinatewise.

We differentiate functions between arbitrary definable sets by "forgetting" the RV-coordinates. More precisely, let

$$f : \mathrm{VF}^n \times \mathrm{RV}^m \longrightarrow \mathrm{VF}^{n'} \times \mathrm{RV}^{m'}$$

be a definable function. For each fixed input RV-tuple t, generic local constancy of definable RV-valued maps [17, Lemma 8.9] implies that, outside a definable set of VF-dimension less than n, the output RV-tuple is constant on a neighborhood of each input a. If this tuple is s, then $\mathrm{dom}(f_{(t,s)})$ contains a neighborhood of a. At such a point (a,t), we define the partial derivatives of f to be those of this local VF-map $f_{(t,s)}$. If $n = n'$ and all the partial derivatives exist, the *Jacobian of f at (a,t)* is the determinant of the matrix whose (j,i)-entry is the ijth partial derivative, and is denoted by $\mathrm{Jcb}_{\mathrm{VF}} f(a,t)$.